Experiment 4 — IC Engine Components

Design of Mechanisms (BE05000341) · Quiz — Short Answers

Q1. Valve-gear mechanism — function and layout in vertical and horizontal engines.

Function: opens and closes the inlet and exhaust valves at the correct crank angles so the cylinder breathes properly. In a 4-stroke engine the camshaft turns at **½ crank speed** (each valve opens once in every two revolutions).

Main parts: cam & camshaft → tappet (follower) → push rod → rocker arm → valve stem → valve; the **valve spring** closes the valve and keeps the follower on the cam.

Vertical engine (OHV): camshaft near the crank, valves in the cylinder head → push rod + rocker arm are used so motion travels upward.

Horizontal engine: cylinder lies on its side; the camshaft and valves lie beside the cylinder, so a short tappet (sometimes a rocker) opens the valve almost directly — push rods may not be needed.

Q2. Numerical — Tubular push rod for an exhaust valve (valve dia = 50 mm, max cyl pressure = 0.4 N/mm², L = 110 mm, E = 2.1 × 10⁵ N/mm², FOS = 4, di/do = 0.75).

Load on push rod $Pp = (\pi/4) \cdot dv^2 \cdot p = (\pi/4) \cdot 50^2 \cdot 0.4 =$ **785 N ≈ 750 N**. Buckling load required $Pcr = Pp \times FOS =$ **3000 N**. Assume both ends hinged, Euler: $Pcr = \pi^2 E I / L^2$. $I = \pi/64 \cdot (do^4 - di^4)$, with $di = 0.75\ do \rightarrow I = 0.0335\ do^4$. $3000 = \pi^2 \times 2.1 \times 10^5 \times 0.0335\ do^4 / 110^2 \rightarrow$ **do ≈ 6.03 mm → select do = 6.5 mm**, $di = 0.75 \times 6.5 \approx$ **4.9 mm**. Rechecked FOS ≈ **4.87 (safe)**.

Ans: Load Pp = **750 N**, buckling load Pcr = **3000 N**.
Push rod: **do = 6.5 mm**, **di ≈ 4.9 mm**, **L = 110 mm**.
Actual FOS ≈ **4.87 (> 4, safe)**.

Q3. Numerical — Valve-gear design for a horizontal gas engine (port 70 mm, lift 25 mm, p_max = 4.5 MPa; valve opens 33° before ODC, closes 1° after IDC; speed 350 rpm; rocker arms 175 mm each at 140°).

Approach. (1) size the **valve** by port area and strength, (2) design the **valve spring** to close the valve and resist suction, (3) design the **rocker lever** for bending and (4) choose the **pins and cam**. Key relations used:

• Valve head thickness $t = k \cdot d_v \sqrt{(p/\sigma b)}$; stem diameter $d_s = 0.25\ d_v + 11$ (approx). • Spring force $F_{min} = p_s \cdot A_v + W_v$; $F_{max} = F_{min} + k \cdot lift$. • Rocker reaction $R = P_{cam} + P_{valve}$; bending at fulcrum $M = P \times arm$; I-section b/d ratio chosen 1 : 2. • Cam angle for lift (from engine timing) = (θ_open) × (camshaft/crank) = (33° + 180° + 1°)/2 = **107°** total; half (rise) = **53.5°**.

Valve. $d_v \approx$ port dia 70 mm → **stem d_s = 15 mm**, **head thickness t = 7 mm**, **lift = 25 mm**.

Spring. Wire **d = 4.5 mm**, mean coil **D = 45 mm**, **10 active (12 total) coils**, free length **118 mm**, forces **140 N (closed) to 254 N (lift)**.

Rocker lever. Design load P ≈ **1202 N**; fulcrum reaction R ≈ **2259 N**; I-section at fulcrum H = **35 mm**, B = **28 mm**, flange t = **7 mm**; each arm **175 mm**.

Pins. Fulcrum pin **14 mm φ × 18 mm**; roller pin **10 mm φ × 13 mm**.

Cam. Rise/return each **53.5°**, lift **25 mm**, base-circle radius **35 mm** (base circle 70 mm with a 30 mm roller) → prime-circle radius **≈ 85 mm**.

Ans: Valve: **stem 15 mm**, **head 7 mm**, **lift 25 mm**.

Spring: wire **4.5 mm**, mean coil **45 mm**, **10 active coils**, free length **118 mm**, force **140 → 254 N**.

Rocker: load **1202 N**, fulcrum reaction **2259 N**, I-section **H = 35, B = 28, t = 7 mm**, arms **175 mm**.

Pins: fulcrum **14 × 18 mm**, roller **10 × 13 mm**.

Cam: rise/return **53.5°**, lift **25 mm**, prime-circle radius **≈ 85 mm**.